

SEDAR 67 Gulf of Mexico Vermilion Snapper

Develop or expand fishery-independent survey coverage to the western Gulf of Mexico and improve age composition sampling of commercial and recreational catch from the western Gulf of Mexico.

Improve sample sizes in the recreational fisheries, particularly for age composition data, so that the recreational fleet can be modeled by mode and/or region.

Given the fast growth and limited age composition information for vermilion snapper, explore the use of a length-based assessment model.

Barring improvement in sampling data from the western Gulf of Mexico, reconsider fleet structure in the assessment to model only a single Gulf-wide unit (i.e., combine data across regions instead of splitting out the western commercial fleet and western headboat CPUE).

Pending improved sampling in the western region, investigate a two-region model that may be better able to account for differences in age structure and recruitment across the Gulf of Mexico.

Continue to evaluate methods to better estimate discards by fleet and attempt to directly fit this data in the assessment model.

Evaluate discard mortality rates and increase the value utilized in the model as appropriate. Further explore the implications of dropping fishery-dependent CPUE indices from the assessment.

Evaluate the protocol for estimating shrimp bycatch and update the WinBugs program with any changes to data collection protocols that may have occurred over the last decade.

Reevaluate the super-year approach for modeling shrimp bycatch to better reflect the low observed bycatch levels in recent years (i.e., using two super-years to reflect the high and low effort regimes pre- and post-2000)

Explore reparametrization of the double normal selectivity curves for the fishery-independent surveys to reduce correlations and improve model stability.

Obtain age or length compositions from the shrimp bycatch fisheries to better inform shrimp selectivity estimates.

Pending expansion of the spatiotemporal coverage of the FWRI repetitive timed drop and vertical longline surveys, explore their use in future assessments.